First Estimates of Food-Web Structure Derived from Species Richness

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1 Introduction

2 Methods

2.1. A maximum entropy model for predicting food-web structure Our mathematical model predicts the structure of species-based food webs from their number of species. Our model involves three main steps: (i) the prediction of the number of links from the number of species using the flexible links model; (ii) the derivation of the joint degree distribution using a maximum entropy approach; and (iii) the prediction of the adjacency matrix using simulations and maximum entropy. The next subsections describe each of these steps in detail.

2.1.1 Predicting the number of links In order to derive the joint degree distribution of maximum entropy, we need two network-level measures: the number of species S and the number of interactions L. While the number of species is a well-described measure of biodiversity for many taxa and locations, the number of interactions is currently difficult to estimate empirically without sampling all pairwise interactions in a biological community. We thus used a predictive statistical model to simulate the number of interactions from the number of species in food webs.

To do so, we used the flexible links model of MacDonald, Banville, and Poisot (2020). The flexible links model incorporates meaningful ecological constraints into the prediction of L, namely the minimum $S-1$ and maximum S^2 numbers of interactions in food webs, and estimates the proportion of the $S^2-(S-1)$ *flexible links* that are realized. More precisely, this model states that the number of *realized* flexible links L_{FL} in a food web represents the number of realized interactions above the minimum (i.e. $L = L_{FL} + S - 1$) and is obtained from a beta-binomial distribution with $S^2 - (S - 1)$ trials and parameters $\alpha = \mu e^{\phi}$ and $\beta = (1 - \mu)e^{\phi}$:

$$L_{FL} \sim \mathrm{BB}(S^2 - (S-1), \mu e^{\phi}, (1-\mu)e^{\phi}), \tag{1}$$

where μ is the average probability across food webs that a flexible link is realized, and ϕ the concentration parameter around μ.

We fitted the flexible links model on all food webs archived on Mangal, an ecological interactions database (Poisot et al. 2016). Ecological networks archived on Mangal are multilayer networks, i.e. networks that describe different types of interactions. We considered as food webs all networks mainly composed of trophic interactions (predation and herbivory types). We estimated the parameters of eq. 1 using a Hamiltonian Monte Carlo sampler with static trajectory (1 chain and 3000 iterations):

$$[\mu, \phi | \mathbf{L}, \mathbf{S}] \propto \prod_{i=1}^{m} \mathrm{BB}(L_i - (S_i - 1) | S_i^2 - (S_i - 1)), \mu e^{\phi}, (1-\mu)e^{\phi}) \times \mathrm{B}(\mu | 3, 7) \times \mathcal{N}(\phi | 3, 0.5), \tag{2}$$

where m is the number of food webs archived on Mangal ($m = 235$) and $\mathbf{L}$ and $\mathbf{S}$ are respectively the vectors of their numbers of interactions and numbers of species. Our weakly-informative prior distributions were chosen following MacDonald, Banville, and Poisot (2020), i.e. a beta distribution for μ and a normal distribution for ϕ. The Monte Carlo sampling of the posterior distribution was conducted using the Julia library `Turing` v0.15.12 (Ge, Xu, and Ghahramani 2018).

The flexible links model is a generative model, i.e. it can generate plausible values of the predicted variable. We thus simulated 1000 values of L for each value of S ranging between 5 and 1000 species using the joint posterior distribution of our model parameters.

2.1.2 Deriving the joint degree distribution The number of species and the number of interactions are state variables whose ratio was directly used to obtain the least-biased degree distribution $p(k)$. This discrete probability distribution represents the probability that a species have k interactions in its food web, k ranging between 1 and S. The least-biased of these distributions, considering our limited knowledge of the system, can be derived using the principle of maximum entropy, i.e. by maximizing Shannon's entropy

$$H = -\sum_{k} p(k) \log p(k) \tag{3}$$

while respecting a set of constraints on the degree distribution. We used two such constraints:

$$g_1 = \sum_{k=1}^{S} p(k) = 1 \tag{4}$$

and

$$g_2 = \langle k \rangle = \sum_{k=1}^{S} k p(k) = \frac{2L}{S} \tag{5}$$

The first constraint g_1 is a normalizing constraint that ensures that all probabilities sum to 1. In addition, the second constraint g_2 fixes the average of the degree distribution to the mean degree $\langle k \rangle$. The mean degree is twice the value of the linkage density L/S since each interaction must

be counted twice when summing all species' degrees. We used each simulated value of L to compute the mean degree constraint. As a result, we derived 1000 maximum entropy degree distributions for each value of S.

Finding probability distributions of maximum entropy is typically done using the method of Lagrange multipliers:

$$\frac{\partial H}{\partial p(k)} = \lambda_1 \frac{\partial g_1}{\partial p(k)} + \lambda_2 \frac{\partial g_2}{\partial p(k)}, \tag{6}$$

where λ_1 and λ_2 are the Lagrange multipliers. Probability distributions of maximum entropy are derived by finding these values. Evaluating the partial derivatives with respect to $p(k)$, we obtained:

$$-\log p(k) - 1 = \lambda_1 + \lambda_2 k \tag{7}$$

Then, solving eq. 7 for $p(k)$, we obtained

$$p(k) = \frac{e^{-\lambda_2 k}}{z}, \tag{8}$$

where $z = e^{1+\lambda_1}$.

After substituting $p(k)$ in eq. 4 and eq. 5, we got a nonlinear system of two equations and two unknowns:

$$\frac{1}{z} \sum_{k=1}^{S} e^{-\lambda_2 k} = 1 \tag{9}$$

$$\frac{1}{z} \sum_{k=1}^{S} k e^{-\lambda_2 k} = \frac{2L}{S}, \tag{10}$$

which implies that $z = \sum_{k=1}^{S} e^{-\lambda_2 k}$.

We solved eq. 10 numerically using the Julia library `NLsolve` v4.5.1 (TK). However, in figure TK, we show how an analytical solution in the limit $S \to \infty$ yields very similar results. For large food webs (TK), the degree distribution of maximum entropy approaches

$$p(k) = c r^k, \tag{11}$$

with

$$c = \frac{1}{\langle k \rangle - 1},$$

and

$$r = \frac{\langle k \rangle - 1}{\langle k \rangle}.$$

Next, the joint degree distribution $p(k_{in}, k_{out})$ of maximum entropy was derived using this analytical solution. The joint degree distribution is the probability of finding a species with k_{in}

predators and k_{out} preys in a food web, with $k = k_{in} + k_{out}$. We first observed that the joint degree distribution is identical to the probability of a species having k_{in} predators and a total of k interactions:

$$p(k_{in}, k_{out}) = p(k_{in}, k) \tag{12}$$

Using Bayes' theorem, we expressed the joint degree distribution as follows:

$$p(k_{in}, k) = p(k_{in}|k)p(k), \tag{13}$$

where $p(k)$ is the degree distribution of maximum entropy derived above. An unbiased way to estimate $p(k_{in}|k)$ is by considering all k interactions as Bernoulli trials with a probability q of 0.5 of being incoming interactions. As a result, the joint probability distribution of maximum entropy of a food web with S species and a total of L interactions is given by:

$$p(k_{in}, k) = \binom{k}{k_{in}} q^{k_{in}}(1-q)^{k-k_{in}} c r^k, \tag{14}$$

which gives, after substitution and simplification:

$$p(k_{in}, k_{out}) = \binom{k_{in} + k_{out}}{k_{in}} c \left(\frac{r}{2}\right)^{k_{in}+k_{out}} \tag{15}$$

2.1.3 Predicting the adjacency matrix The number of species S, the number of interactions L, and the joint degree distribution $p(k_{in}, k_{out})$ are further constraints on the adjacency matrix of food webs. Directly deriving a network of maximum entropy considering these constraints is beyond our mathematical abilities. However, we worked around this issue by first simulating many networks of S species whose joint degree distribution was identical to the one obtained above. We next measured the entropy of all these potential networks and kept the one with maximum entropy.

Network simulations were conducted in two steps. First, we simulated the number of species among S species with k_{in} in-degrees and k_{out} out-degrees using the joint probability distribution (TK). We removed all occurrences with a total number of interactions different from L. Next, we generated *a maximum* of 1000 random Boolean matrices that maintain the numbers of in and out-degrees for all species in the food web. To do so, we used the curveball algorithm of Strona et al. (2014) implemented in the Julia library `RandomBooleanMatrices` v0.1.1. In order to respect biological constraints, we removed all occurrences with species having no interactions

We then computed the entropy of all simulated networks using an entropy measure for binary matrices (TK). We kept the network with the greatest entropy value. As a result, for each value of S we obtained 1000 networks of maximum entropy having different predicted total numbers of interactions L.

2.2. Worldwide spatial variation of food-web structure A main application of our model is the spatial analysis of food-web structure at very large spatial scales. We used data from BiodiversityMapping to estimate the number of terrestrial vertebrate species (mammals, birds, and amphibians) in 10 km x 10 km grids worldwide. For illustrative purposes, we considered that all species present in a grid were part of the same food web and interacted with at least one species in their community.

We predicted the structure of all food webs using our model. We first generated networks of maximum entropy for the range of species richness encountered in the dataset (TK). We then computed a set of emerging properties (i.e. nestedness, maximum trophic level, network diameter, and entropy) for all of these networks, and mapped the spatial variation of these measures according to the number of species in each grid. In figure TK, we present the structure of networks having a number of interactions L of maximum a posteriori (MAP) probability.

2.3. Data and code availability All code and data to reproduce this article are available at the Open Science Framework (TK). Our analyses and simulations were conducted in Julia v1.5.4.

3 Results

Anticipated figures: - Conceptual figure (maybe) - Numerical solutions as a function of analytical solutions for the maximum entropy degree distributions - Probability distribution of mean-degree constraints for a food-web of S species (maybe) - Maximum entropy distributions of a food web of S species and different numbers of links L - Maxent against empirical degree distributions - Predicted as a function of empirical measures of network structure - Predicted and empirical measures of network structure as a function of species richness - 4 worldwide maps of predicted network properties (nestedness, maximum trophic level, network diameter, entropy)

4 Discussion

5 Conclusions

6 Acknowledgments

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We wrote this article on land located within the traditional unceded territory of the Saint Lawrence Iroquoian, Anishinabewaki, Mohawk, Huron-Wendat, and Omàmiwininiwak nations.

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